AIMScribe v3 — The Complete Cloud Plan

Full cloud and partial cloud, DigitalOcean and Amazon, priced monthly and yearly · Storage on Cloudflare R2 · AIMS LAB · 5 October 2026

This plan assumes	Value
Consultations in the collection phase	18,000, over about 2.5 months
Consultations per clinic day	277
Average recording	120 MB — 24 minutes at 44.1 kHz, mono, 16-bit
Audio storage you will buy	**3 TB**
Storage provider	**Cloudflare R2** — your policy
Running	permanently, not only for the study

1 · The words, before the numbers

A rented computer is sold by three numbers, and a storage service by two more. None of them are explained anywhere you buy them, so here they are.

Word	What it really means	Think of it as
vCPU ("cores", "workers")	How many jobs the machine can do in the same instant. 4 vCPU means four things happen side by side instead of waiting in line	**How many staff you employ.** One person does one job at a time
GB RAM ("memory")	The space those staff have to work in. A program must be loaded into memory before it can run. Run out and the machine crawls, then stops	**The size of the desk.** Too small and the work falls on the floor
GB disk ("SSD", "storage")	Where things are kept when nothing is using them: the operating system, the databases, files waiting their turn	**The filing cabinet.** It holds things; it does no work
Object storage (R2, S3, Spaces)	A storage service you reach over the network, not a disk. Unlimited, cheap, replicated by the provider. You cannot run a database on it, but it is ideal for finished files	**A warehouse** down the road. Enormous, cheap, slightly further away
Egress ("data transfer out")	What a provider charges to send your own data out of their network. **This is the cost that surprises people**	**A toll on the road out.** Free at some gates, expensive at others

So **"8 vCPU / 16 GB / 320 GB"** — DigitalOcean's **s-8vcpu-16gb** — is one rented computer: eight staff, a desk big enough for them, and a 320 GB filing cabinet. Section 5 explains each of those three numbers.

2 · What goes into the 3 TB, and how long it lasts

The format decides everything else, so it comes first. **On a full-cloud deployment we keep the original WAV and convert nothing.** There is no FLAC step and no cold archive tier: the recording that comes off the microphone is the recording that is stored, permanently, in Cloudflare R2.

What you store	Per recording	The 18,000 study	What 3 TB holds	Clinic days of recording
Raw WAV — what we do	**120 MB**	**2.06 TB**	**26,200 recordings**	**95 days, about 3.6 months**
FLAC, if we compressed	72 MB	1.24 TB	43,700 recordings	158 days

Store the raw WAV. 3 TB holds 26,200 consultations and the whole 18,000-patient study needs 2.06 TB, which leaves about 0.94 TB spare — roughly another month of recording. Compressing to FLAC would roughly double that headroom, but it buys capacity we do not need and costs a conversion step, a verification step and a second format to keep track of.

This is the decision that follows from going fully cloud. FLAC earned its place when the plan was a 100 GB server at UIU feeding a cold archive, because every gigabyte on that disk was scarce. With storage rented by the gigabyte and no archive tier, the saving is $21 a month and the cost is a more complicated pipeline on the one part of the system that must never lose anything.

No Glacier, no deep archive. Deep archive is cheap because reading from it takes 12 to 48 hours. Your researchers will read this dataset constantly for the next year, so everything stays in normal storage where it opens instantly. Revisit this in a few years, when old recordings genuinely are not being touched — not now.

When 3 TB fills, you choose	What it means
Buy more	R2 has no limit; it is $15 per TB per month. Another 3 TB is $45
Move the oldest months to deep archive	Cheapest, and only sensible once nobody is reading them
Delete the oldest months	Only if the research is finished. Nothing else holds a copy by then

3 · Storage: Cloudflare R2 — your policy, costed

Item	Rate	At 3 TB	Monthly	Yearly
Storage	$0.015 per GB	3,000 GB	**$45**	**$540**
Uploading files (writes)	$4.50 per million	~22,000/month	under $1	~$2
Reading files (reads)	$0.36 per million	research use	under $1	~$3
Sending data out (egress)	**$0 — always**	any amount	**$0**	**$0**
Total			**~$46**	**~$545**

Why R2 is the right policy choice, in one sentence: every other provider charges you to read your own data back, and you are going to read this dataset many times over.

Reading the full 3 TB once would cost about **$250 on Amazon S3** and about **$30 on DigitalOcean Spaces**. On R2 it costs **nothing**, however often you do it. For a research dataset that will be downloaded for transcription, for model training, and again every time somebody re-runs an experiment, that is the difference between a fixed bill and a growing one.

4 · The trap that decides which server to rent

R2 never charges you to take data **out**. But something has to put data **in** — and that is your server, sending about **844 GB every month** up to Cloudflare. Whether that is free depends entirely on who you rent the server from.

Your server is at	What it charges to send 844 GB/month out	Monthly	Yearly
DigitalOcean	The droplet includes **5,000 GB** of transfer a month. 519 is well inside it	**$0**	**$0**
Amazon EC2	First 100 GB free, then **$0.1093 per GB** to leave AWS. 419 GB is billable	**$46**	**$549**
Amazon EC2 → Amazon S3	Free, if you create a VPC gateway endpoint (free, but you must set it up)	$0	$0

So R2 and Amazon are a poor pair, and R2 and DigitalOcean are a good one. Choosing Amazon and R2 together costs you $549 a year for nothing — the data is simply walking out of Amazon's gate. If the storage policy is R2, the sensible server is DigitalOcean. If the server must be Amazon, the sensible storage is S3.

This is the single most important sentence in this document, and it is the kind of thing no pricing page tells you.

5 · The server, number by number

Staff (vCPU): the work needs 1.5; the droplet has 8

Job	How much work it really is	Staff needed
Uploading the finished recording	Reading one file and sending it. No conversion, so almost no processor time at all	under 1
The website, the API, CMED's messages	About 1.2 requests a second across all 14 rooms, measured. That is very light	well under 1
Both databases	10 GB of data, simple lookups	well under 1
Queue, gateway, housekeeping	Background work	negligible
Total		**about 1.5**

The measured work needs about 1.5; eight is AIMS LAB's requirement (at least 8 cores, 8 GB and 200 GB), and on DigitalOcean the only regular droplet that meets all three is **s-8vcpu-16gb**. The 4-vCPU size has a 160 GB disk, below the 200 GB asked for. The spare staff are not wasted: they absorb the busy hour, when fourteen rooms finish at once and fourteen recordings are joined, encrypted and uploaded together, and they leave room to switch transcription on.

Desk (memory): 16 GB, and how it is shared

Part	Normal use	Its cap on the droplet
PostgreSQL, both databases	240 MB + its 4 GB cache	6 GB
The API server	320 MB	2 GB
Archive worker	250 MB	2 GB — joins one long consultation
Transcription, if switched on	—	1 GB
Backup job	a few minutes a night	512 MB
Queue, gateway, monitor, pooler	about 150 MB	about 900 MB
Everything together	**about 1 GB, plus the cache**	**about 12.4 GB**, leaving 3.6 GB for the system

Normal use was measured under a test twice as heavy as the real clinics, and is about 1 GB. The caps are ceilings, not reservations: each one stops a single part from taking the machine down. The database's cache is set to 4 GB (`postgresql.16gb.conf`), a quarter of the droplet, which keeps the whole database in memory for years. 8 GB would run the measured load; 16 GB is what the required 8-core droplet comes with, and it is used.

6 · Filing cabinet (disk): 320 GB, and no extra disk

The audio does not stay on the server for long. Each recording is encrypted and uploaded to R2 as the raw WAV; once R2 has confirmed it holds exactly what was sent, the WAV is kept on the server for **2 days** for quick re-reading and then removed. Its clinical JSON, manifest and index entry stay. So the disk holds what must physically live on the machine, plus two days of recordings:

What	Size	Why it has to be there
Operating system	20 GB	Linux, packages, updates
The programs	15 GB	Eight parts: the API, PostgreSQL, Redis, the archive worker, the gateway, the connection pooler, the viewer, CMED's page
Both databases	**10 GB**	Patient records, prescriptions, previous visits, and the tracking record of every audio file. **Grows 20–40 GB a year**
Database safety log	10 GB	Every change written down before it is applied, so a power cut cannot corrupt anything
Backup staging	10 GB	Last night's backup, before and after upload
Working space	3 GB	One consultation being joined and encrypted
Upload buffer	**30 GB**	**If R2 is unreachable, finished recordings wait here instead of being lost.** 30 GB is about a day and a half of recording
Logs	5 GB	What each part did, for diagnosing problems
Kept free deliberately	20 GB	The worker refuses to write below this, so a full disk can never damage a recording
Never-full headroom	25 GB	No disk should run above 80% full
Recent recordings	**66 GB**	**Two clinic days at 33 GB a day** (277 consultations × 120 MB). Set by `AIMS_ARCHIVE_LOCAL_RETENTION_DAYS=2`. Nothing is removed until R2 has confirmed its copy
Total needed	**214 GB**	67% of the 320 GB disk

The s-8vcpu-16gb droplet includes 320 GB. 214 fits inside it with a third to spare, so no extra disk is needed and none is budgeted. On Amazon the disk is billed separately — a real difference between the two, not a preference.

Two lines to watch over the years. The databases grow 20–40 GB a year, which the spare 100 GB covers for about three years. And if R2 stops confirming copies, recordings stay on the server and the disk rises: the server raises `cloud_copy_overdue` for any recording without a copy after a day.

7 · Full cloud — DigitalOcean

Everything rented. Nothing at UIU except the laptops in the clinics.

What to buy	Exactly what to pick	Monthly	Yearly
The server	Droplet → Basic → Regular. **s-8vcpu-16gb: 8 vCPU / 16 GB / 320 GB SSD**. Region: **Bangalore (BLR1)**	$96	$1,152
Server backups	Tick "Enable backups" when creating it — 20% of the droplet price	$19.20	$230
Audio storage	**Cloudflare R2**, 3 TB, bought from Cloudflare separately	$45	$540

Domain name	e.g. aimscribe.uiu.ac.bd — or use a UIU subdomain for free	~$1	~$12
HTTPS certificate	Free, renews itself	$0	$0
Database	**Nothing to buy.** PostgreSQL runs on the server itself	$0	$0
Sending data to R2	Included in the droplet's 5 TB transfer	**$0**	**$0**
TOTAL		**$161**	**$1,934**

For the 2.5-month collection phase: about $350, because storage is only half full on average while it fills.

8 · Full cloud — Amazon Web Services

Use the **Mumbai region (ap-south-1)** — nearest to Dhaka, about 40 milliseconds away. Two versions are priced: with R2 as your policy states, and with S3 which avoids the transfer charge.

What to buy	Exactly what to pick	Monthly	Yearly
The server	EC2 → **t4g.large** (2 vCPU, 8 GB, ARM). **t4g.xlarge** (4 vCPU, 16 GB) if you want the headroom DigitalOcean gives by default	$49 *(xlarge $98)*	$588 *($1,176)*
The disk	**EBS gp3 volume, 200 GB.** On Amazon this is a separate purchase — it does not come with the server	$18	$216
Backups	EBS snapshots, daily, keep 7 days	$10	$120
Fixed address	Elastic IP — free while attached to a running server	$0	$0
Database	**Nothing to buy.** PostgreSQL runs on the server itself	$0 *(RDS ~$25)*	$0 *($300)*
Audio storage — option 1	**Cloudflare R2**, 3 TB, per your policy	$45	$540
… plus transfer out	**844 GB/month leaving AWS at $0.1093/GB**	**$46**	**$549**
TOTAL with R2		**$168**	**$2,013**
Audio storage — option 2	Amazon S3 Standard, 3 TB, same region	$75	$900
TOTAL with S3	**transfer free via a VPC gateway endpoint**	**$152**	**$1,824**

Three things you must set up on Amazon that DigitalOcean does for you

	What	Why it matters
1	A **VPC gateway endpoint** for S3 (free)	Without it, traffic between your own server and your own S3 bucket can be billed as if it left Amazon. This single setting is the difference between $0 and about $46 a month
2	A **billing alarm** at, say, $250	Amazon will let a misconfiguration run for a month before you see it. DigitalOcean's prices are mostly fixed; Amazon's are mostly per-use
3	**Versioning** on the storage bucket	So a mistaken delete can be undone. It costs almost nothing at this size

9 · Partial cloud — your own server, storage in the cloud

The clinics and the audio work exactly the same. The difference is that the computer running the software sits at UIU instead of being rented.

What	Detail	Monthly	Yearly
The server	The 8-core / 8 GB machine AIMS LAB offers	$0	$0
Its disk	**100 GB is not enough — 160 GB is needed.** A 256 GB SSD is a one-off purchase of roughly $30	—	**~$30 once**
Audio storage	Cloudflare R2, 3 TB	$45	$540
Domain name	A UIU subdomain	~$1	~$12
TOTAL		**$46**	**$552**

What you gain	What you take on
Saves about $1,380 a year against full cloud	UIU's network rules must allow it: a fixed name, one port open, no filtering of the connection to Cloudflare
Patient data stays on a university machine, which simplifies the ethics question	Power cuts, cooling and the physical machine become your problem
No monthly server bill, ever	Nobody else is on call. If it stops at 11pm, somebody at AIMS LAB drives in
Fast local access for researchers on campus	Backups, patching and disk monitoring are all yours to do

The honest summary: partial cloud is cheaper by about $115 a month, and the saving is real. What you pay instead is in attention and availability — somebody has to own that machine. For a 2.5-month study with staff on hand, that is a fair trade. For permanent running with nobody specifically assigned to it, the rented server is worth the money.

10 · Partial cloud — the full specification

Section 9 priced it. This is what UIU actually has to provide, to the same level of detail as the rented server in sections 5 and 6. Nothing about the clinics, the laptops or the storage changes — only where the computer sits.

10.1 The machine

	Minimum	Recommended	What AIMS LAB offers	Verdict
Staff (cores)	4	8	**8**	**enough**
Desk (memory)	8 GB	16 GB	**8 GB**	**enough**
Filing cabinet (disk)	**160 GB SSD**	**256 GB SSD**	**100 GB**	**too small**
Network card	1 Gbit/s	1 Gbit/s	assumed	check
Power protection	UPS	UPS with clean shutdown	unknown	check

Only the disk is wrong, and only by 60 GB. Section 6 works out why 148 GB is needed even with all the audio in R2 (set `AIMS_ARCHIVE_LOCAL_RETENTION_DAYS=0` there, so each WAV leaves as soon as its copy is confirmed). A 256 GB SSD is a one-off purchase of roughly $30 and solves it with room for three years of database growth. Nothing else about the machine needs changing.

10.2 Software on it

What	Which	Why
Operating system	**Ubuntu Server 24.04 LTS**	Supported until 2029. On an 8 GB machine, Windows plus Docker Desktop would consume about half the memory before AIMScribe started
Container engine	Docker Engine with the Compose plugin	The whole system is nine services started by one command
Access	root or sudo on the host	To install and to run the stack
Everything else	nothing to install by hand	Both databases, their roles and all fifteen schema scripts are built on first start

10.3 What UIU's network team must agree

This is the part that takes longest, because it is other people's decision. A rented server needs none of it.

	Requirement	Why it cannot be skipped
1	**Inbound port 443**, TCP **and** UDP	Every clinic laptop and CMED's server connect in on it. UDP carries HTTP/3
2	**Inbound port 80**, TCP	Certificate renewal only. It serves nothing and redirects to 443
3	SSH inbound from **named UIU machines only**	Administration
4	**Outbound 443 to Cloudflare, unfiltered**	This is how recordings reach R2. **No TLS interception and no transparent proxy** — uploads are signed with the destination inside them, and a middlebox breaks the signature
5	**Outbound NTP (port 123)**	Recordings are filed under the clinic's local date, decided once. A drifting clock misfiles consultations permanently
6	**A DNS name, lower case, and a static address**	Certificates are issued for names. A changed DHCP lease silenced every recorder once already on the bench
7	No aggressive idle-connection timeout	Recorders reuse a connection every 30 seconds; the server holds them 75

8	**No automatic reboots or patch windows 08:00–21:30**, six days a week	A reboot mid-consultation is survivable but costs recordings that cannot be re-taken
9	The data disk exempt from quota sweeps and automatic cleanup	It holds the databases and the upload buffer

If CITS will not open an inbound port at all, there is still a way: a Cloudflare Tunnel needs **no inbound rule whatsoever** — the server dials out and the tunnel carries traffic back. It is free, it fits the R2 policy, and it removes requirements 1, 2 and 6 from the list above.

10.4 One-off costs, and who owns the machine

Item	Cost	Note
256 GB SSD	~$30	Required. 100 GB is not enough
UPS, if the room has none	$80–150	A recording being written during an unclean power cut can be damaged
One-off total	**$30–180**	
Then, per year	**$552**	R2 storage and the domain only

The cost that does not appear on any invoice is a person. With a rented server, backups are a checkbox, the hardware is somebody else's problem, and a failed disk is replaced without you knowing. On your own machine, all of that is a named person at AIMS LAB:

Job	How often	What happens if nobody does it
Check the backup ran, and that the dashboard is clean	every morning, 2 minutes	A silent failure runs for weeks
Restore one recording and prove it opens	day one, then monthly	You discover the backups were unreadable on the day you need them
Operating system security updates	monthly, 20 minutes	A public-facing machine drifts out of support
Watch free disk space	weekly glance at the dashboard	The upload buffer fills and archiving stops
Be reachable if it stops	whenever it happens	Clinics stop recording after the laptops fill, about 13 hours

10.5 The two side by side

	Partial — your own machine	Full — DigitalOcean
Yearly cost	**$552**	$1,934
One-off cost	$30–180	$0
Time to running	**weeks** — network approvals first	**an afternoon**
Backups	you build, you test	a checkbox, then you test
Hardware failure	your problem, your spare parts	theirs; you rebuild from a snapshot
Someone on call	a named person at AIMS LAB	the provider, for the machine
Network rules	nine agreements with CITS	none
Patient data location	**on a UIU machine**	on a rented machine in Bangalore
Power cuts, cooling	yours	theirs
If the project grows	buy and install hardware	change a size in a menu

The honest way to choose. Partial cloud saves about $1,380 a year. That is roughly one week of a developer's time. If AIMS LAB has somebody who will genuinely own that machine — the morning check, the monthly restore, the 11pm phone call — take it, and keep patient data on a university machine into the bargain. **If that person does not exist and cannot be named today, rent the server**: $1,380 a year is cheap compared with discovering in month three that nothing has been backed up.

11 · Everything side by side

Setup	Server	Storage	Transfer	Monthly	Yearly
Partial — own server + R2	$0	$45	$0	**$46**	**$552**
Full — DigitalOcean + R2	$115	$45	$0	**$161**	**$1,934**
Full — DigitalOcean + Spaces	$115	$60	$0	$176	$2,112
Full — Amazon + S3	$77	$75	$0	$152	$1,824
Full — Amazon + R2	$77	$45	**$46**	$168	$2,013

If the storage policy is Cloudflare R2, there are only two sensible answers: your own server (cheapest, if somebody owns it), or DigitalOcean (nobody owns it). The Amazon rows are priced at a smaller server (2 vCPU, 8 GB); at the required 8 cores Amazon costs more than DigitalOcean, not less. **Amazon with R2 is the worst of all five** — you pay Amazon $549 a year for the privilege of sending your data to Cloudflare.

12 · What runs where

Layer	What	Lives on	If it stops
Clinic	AIMScribe agent, one per room	the doctor's laptop	That room stops. The other thirteen continue
Clinic	CMED's page	the doctor's laptop, in a browser	No new consultation can be opened
Server	The API: permissions, CMED's messages, the dashboard	the server	Laptops keep recording and hold up to 13 hours each
Server	**aims_recordings** — audio tracking	the server's disk	Nothing can be confirmed or archived
Server	**aims_clinical** — patients, prescriptions	the server's disk	Recording continues and is archived; CMED's records attach when it is back
Server	Archive worker — joins, encrypts, uploads	the server	Recordings queue in the 30 GB buffer
Storage	The recordings, as WAV	Cloudflare R2	Finished recordings wait in the buffer; clinics unaffected for ~1.5 days

13 · Keeping it running

Risk	What already protects you	What you must still do
Server down an hour	Laptops hold ~13 hours each. Nothing is lost	Nothing. This is designed for
Server down a day	Laptops begin to fill; clinics eventually stop	**Alerting**, so it is noticed in minutes not at end of shift
R2 unreachable	The 30 GB buffer absorbs about a day and a half	An alert when the buffer passes half full
Disk fills	The worker refuses to write rather than corrupt anything	Watch the dashboard; it reports this
A recording CMED never describes	Kept and archived, with no clinical record — never deleted, but silent	Check the unconfirmed count every morning. Once any outage is over it should be zero; ask CMED to check its outbox for any left
The databases are lost	Nightly encrypted backup to R2	**Restore one on day one, then monthly.** A backup nobody has read is not a backup
Someone deletes audio by mistake	Nothing, by default	Turn on a bucket lock (retention rule) on the archive bucket, if Cloudflare offers it
The bill surprises you	R2 is a fixed rate with no egress	On Amazon only: set a billing alarm

One server, not two. What keeps clinics recording through an outage is not a second server — it is the buffer on each laptop, which you already have. A second server doubles the cost and more than doubles the complexity. Spend the effort on alerting and on testing restores.

14 · What had to be built, and is

The system used to write finished recordings to the server's own disk and never remove them, which fills any disk in about a week. That is fixed; the order is the one this plan asked for.

	Work	Now
1	**Upload the finished recording to R2, verify it arrived, then remove it from the server**	Done. R2's own fingerprint of the object must match what was sent before the copy counts; the WAV leaves after `AIMS_ARCHIVE_LOCAL_RETENTION_DAYS`, and only then
2	**No conversion step** — upload the WAV as it is	Done. The copy is the raw WAV, encrypted, decrypted again and hashed before upload
3	Fetch a recording back from R2	`restore.py` turns a downloaded copy back into the WAV, byte for byte; the manifest beside it names the object
4	Alert when copies stop arriving	`cloud_copy_overdue` after a day without a copy

The order is the safety: upload, verify, *then* remove. Nothing is removed on the strength of an upload that was merely accepted, and nothing at all while no copy key is set.

15 · Setting it up, in order

	Step	Where
1	Open a Cloudflare account and create the two R2 buckets	Cloudflare
2	Create the server: s-8vcpu-16gb (8 vCPU / 16 GB / 320 GB), Bangalore, backups ticked	DigitalOcean
3	Point a domain name at it, and let the certificate issue itself	DNS
4	Install Docker, copy the project, fill in the settings file	the server
5	Start everything with one command; all fifteen schema scripts run themselves	the server
6	Register the seven clinics and issue CMED's key	the server
7	Run the load test: 28 rooms at once. It must lose nothing	your laptop
8	Install the recorder on one clinic PC and record one real consultation	a clinic
9	**Restore one recording from R2 and prove it opens**	your laptop
10	Turn on alerting, then open the clinics	—

16 · One page to remember

Question	Answer
What does "8 vCPU / 16 GB / 320 GB" mean?	One rented computer, DigitalOcean's s-8vcpu-16gb: eight staff, a desk for them, a 320 GB filing cabinet
Why 8 staff?	AIMS LAB's requirement. The measured work needs 1.5; the rest covers the busy hour and transcription
Why 320 GB and no extra disk?	Audio lives in R2 and stays here only 2 days after its copy is confirmed, so the server needs 214 GB — and 320 comes included
Where do we buy storage?	**Cloudflare R2**, 3 TB — $45/month, $540/year, and nothing to read it back
Do we keep WAV or FLAC?	**Raw WAV.** No conversion. 3 TB holds 26,200 consultations; the study needs 2.06 TB
Do we need Glacier?	**No.** Everything stays instantly readable
Cheapest overall	**Partial cloud — $46/month, $552/year**, if somebody at AIMS LAB owns the machine
Best rented option	**DigitalOcean + R2 — $161/month, $1,934/year**
Amazon?	$152/month with S3. **$168 with R2 — avoid**, Amazon charges $549/year to send data to Cloudflare
How long does 3 TB last?	The whole 18,000 study uses 2.06 TB. 3 TB is about 95 clinic days of recording
What had to be built first?	Upload to R2, verify, then remove locally. Done
The one step never to skip	**Restore a recording on day one and prove it opens**

Prepared by AIMS LAB. Prices are indicative, October 2026, and should be confirmed on each provider's own pricing page before purchase. Server and storage figures are measured on this project's own test bench.